

Program 8 – Execution Steps

Step 1: Create project directory

```
mkdir Program8  
cd Program8
```

Step 2: Create server.js

nano [server.js](#)

```
const http = require('http');  
const port = 3000;  
  
const server = http.createServer((req, res) => {  
  res.end("Hello from Node.js running inside Kubernetes!!");  
});  
  
server.listen(port, () => {  
  console.log(`Server running at port ${port}`);  
});
```

Step 3: Create ConfigMap YAML

nano nodejs-configmap.yml

```
apiVersion: v1  
kind: ConfigMap  
metadata:  
  name: nodejs-app-config  
data:  
  server.js: |  
    const http = require('http');  
    const port = 3000;  
  
    const server = http.createServer((req, res) => {  
      res.end("Hello from Node.js running inside Kubernetes!!");  
    });  
  
    server.listen(port, () => {  
      console.log(`Server running at port ${port}`);  
    });
```

Step 4: Create Pod YAML

nano nodejs-pod.yml

```

apiVersion: v1
kind: Pod
metadata:
  name: nodejs-pod
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs
      image: node:14-alpine      # lighter image
      workingDir: /usr/src/app
      command: ["node", "server.js"]
      ports:
        - containerPort: 3000
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config

```

Step 5: Apply the ConfigMap and Pod

kubectl apply -f nodejs-configmap.yaml

kubectl apply -f nodejs-pod.yaml

Check pod status

kubectl get pods

```

1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Gaming-3-15IMH05:~/prog8$ kubectl apply -f nodejs-configmap.yaml
kubectl apply -f nodejs-pod.yaml
configmap/nodejs-app-config unchanged
pod/nodejs-pod created
1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Gaming-3-15IMH05:~/prog8$ kubectl get pods

```

NAME	READY	STATUS	RESTARTS	AGE
demo-pod	1/1	Running	1 (7h31m ago)	8h
labeled-pod	1/1	Running	1 (7h31m ago)	7h52m
nginx-deployment-6f9664446b-sckkm	1/1	Running	0	66m
nginx-deployment-6f9664446b-tj72l	1/1	Running	0	66m
nodejs-pod	1/1	Running	0	17s
simple-pod	1/1	Running	1 (7h31m ago)	7h59m

Step 8: Access the Node.js application

kubectl port-forward pod/nodejs-pod 3000:3000

Step 9: Open browser

<http://localhost:3000>

